WORKED SOLUTION

STEP I Specimen paper, Question 10

Mechanics • Power and Resistance

STEP I Specimen paper, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

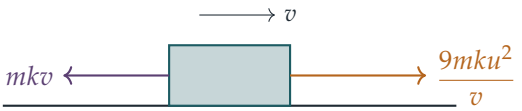
step.maths.org/questions/spec-s1-q10

The key idea

Power is force times speed, so an engine working at constant power P pulls with a driving force P/v. Newton's second law then gives a first-order differential equation for v that separates. Choosing the right form of acceleration is what matters: $\mathrm{d}v/\mathrm{d}t$ when the question asks about *time*, and $v\,\mathrm{d}v/\mathrm{d}x$ when it asks about *distance*.

Solution

Model the engine as a particle of mass m on a level track, so its weight and the normal reaction cancel and only horizontal forces matter. Take the direction of motion as positive.



The maximum speed

At speed v the driving force is $9mku^2/v$, so Newton's second law gives

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{9mku^2}{v} - mkv \quad\Longrightarrow\quad \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{k\,(9u^2 - v^2)}{v}.$$

While $v < 3u$ the right-hand side is positive, so the engine speeds up. It would stop accelerating only where the driving force balances the resistance, $9u^2 = v^2$. So the maximum possible speed is

$$v_{\max} = 3u.$$

Ask yourself The engine always accelerates while $v < 3u$, so why doesn't it reach $3u$?

The acceleration shrinks to zero as $v \to 3u$. Separating the variables, the time to go from speed v_1 to speed v_2 is

$$t = \frac{1}{k}\int_{v_1}^{v_2} \frac{v}{9u^2 - v^2}\,\mathrm{d}v = \frac{1}{2k}\ln\left(\frac{9u^2 - v_1^2}{9u^2 - v_2^2}\right),$$

and this tends to infinity as $v_2 \to 3u$. So $3u$ is a terminal speed: the engine gets as close to it as we like, but never actually reaches it in a finite time. (Equivalently, $v = 3u$ is an equilibrium of the differential equation, and by uniqueness of solutions a solution starting below it can't reach it.)

From u to $2u$

Put $v_1 = u$ and $v_2 = 2u$ in the formula above:

$$t = \frac{1}{2k}\ln\left(\frac{9u^2 - u^2}{9u^2 - 4u^2}\right) = \frac{1}{2k}\ln\left(\frac{8u^2}{5u^2}\right) = \frac{1}{2k}\ln\left(\frac{8}{5}\right),$$

as required.

Power cut off

Now the only forces are the two resistances, mkv and mkB, both opposing the motion. We want a distance, so write the acceleration as $v\,\mathrm{d}v/\mathrm{d}x$:

$$mv\frac{\mathrm{d}v}{\mathrm{d}x} = -mk(v + B) \quad\Longrightarrow\quad \frac{v}{v+B}\,\mathrm{d}v = -k\,\mathrm{d}x.$$

Writing $\dfrac{v}{v+B} = 1 - \dfrac{B}{v+B}$ and integrating as v falls from $2u$ to 0 while x increases from 0 to D:

$$\begin{aligned} kD &= \int_0^{2u}\left(1 - \frac{B}{v+B}\right)\mathrm{d}v \\ &= \Big[v - B\ln(v+B)\Big]_0^{2u} \\ &= 2u - B\ln\left(\frac{2u+B}{B}\right). \end{aligned}$$

So the distance travelled with the power cut off is

$$D = \frac{1}{k}\left[2u - B\ln\left(1 + \frac{2u}{B}\right)\right].$$

This time the engine does stop in a finite time: $\mathrm{d}v/\mathrm{d}t = -k(v+B) \le -kB < 0$, so the deceleration never drops below kB. That's exactly why the extra constant resistance is needed.

Checks

- As $B \to 0$, $B\ln(1 + 2u/B) \to 0$, so $D \to 2u/k$. That matches the resistance mkv alone, where $v\,\mathrm{d}v/\mathrm{d}x = -kv$ gives v falling linearly with distance, at rate k.

- $D > 0$, since $\ln(1+y) < y$ for $y > 0$ gives $B\ln(1+2u/B) < 2u$. And D decreases as B grows, as it should.
- Numerically, with $k = u = B = 1$, stepping the equations of motion gives a stopping distance of 0.90139, matching $2 - \ln 3 \approx 0.90139$. Stepping the powered phase for time $\frac{1}{2}\ln\frac{8}{5} \approx 0.2350$ from $v = 1$ ends at $v = 2.0000$.

Common slips

- Treating the driving force as constant. At constant power, the force is P/v and falls as the engine speeds up.
- Using $\mathrm{d}v/\mathrm{d}t$ in the last part. That gives the stopping time, not the distance.
- Claiming the engine reaches $3u$ "eventually". The time integral diverges, so it never does.

Beyond STEP

The speed $3u$ is a stable equilibrium of the autonomous equation $\dot{v} = f(v)$. Solutions approach it exponentially but never meet it, because solutions of a well-behaved differential equation can't cross or merge. That's a first taste of the uniqueness theorem and of phase-line analysis in a first course on differential equations. Boyce and DiPrima, *Elementary Differential Equations*, covers both. For more on power and resistive forces, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

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